

**İTÜ ELECTRICAL ENGINEERING
SYLLABUS**

Instructor: Asst. Prof. Fatih Küçüktezcan, Room: 7207, e-mail: kucuktezca@itu.edu.tr

Dersin Adı				Course Name		
Elektrik Şebekeleri İçin Bilgisayar Ağları				Computer Networks for Power Systems		
Kodu (Code)	Yarıyılı (Semester)	Kredisi (Local Credits)	AKTS Kredisi (ECTS Credits)	(Course Implementation, Hours/Week)		
				Ders (Theoretical)	Uygulama (Tutorial)	Laboratuvar (Laboratory)
ELK455E	7	2,5	5	2	1	0
Bölüm / Program Department/Program		Elektrik Mühendisliği (Electrical Engineering)				
Dersin Türü (Course Type)		Zorunlu (Required)		Dersin Dili (Course Language)		İngilizce (English)
Dersin Önkoşulları (Course Prerequisites)		-				
Course Category by Content, %		Temel Bilim (Basic Sciences)	Temel Mühendislik (Engineering Science)	Mühendislik Tasarım (Engineering Design)	İnsan ve Toplum Bilim (General Education)	
		-	20	80	-	
Course Description		Smart Grids and communications requirements. Basics of communications engineering. Computer networks. OSI reference and Internet communications models. Transmission media. Communications protocol for power system applications. Transmission and network switching. Intelligent electronic devices (IED). Real-time networks for power systems.				
Course Objective		Communications requirements for power systems. • Communications requirements for power systems. • Computer systems and communication networks. • OSI and Internet Communication Models • Dominant network protocols and technologies • Computer networking for power systems.				
Course Learning Outcomes		i. A knowledge of basic of computer systems. ii. A knowledge of basic communications engineering. iii. A knowledge computer networking. iv. A knowledge of network protocols applied to power systems. v. An ability to design computer networks for power system applications.				
Textbook		Fitz Gerald; “Business Data Communications and Networking” , 14th Edition, John AlanDennis, Wiley & Sons, ISBN: 978-0-470-05575-5, 2020.				
(Other References)		B Forouzan, “Data communication and networking”, Fourth addition. July, 2010. 1. A S Tanenbaum, “Computer Networks”, 4th Edition, Pearson Education Ltd. NJ. 2003 2. H Hrasnica, A Haidine, R Lehnert, “Broadband Powerline Communications - Network Design”, Wiley Ltd. 2004.				
Homework & Projects						
(Computer Use)						
(Other Activities)						

(Assessment Criteria)	(Activities)	(Quantity)	(Effects on Grading, %)
	Midterm Exams	2	25+25
	Short Exams	1-3	10
	Final Exam	1	40

COURSE PLAN

Weeks	Topics	Outcomes
1	An Introduction to Computer Systems.	i, ii, iii
2	Power system communications requirements.	i, ii, iii
3	Basics of communications Engineering.	i, ii, iii
4	An introduction to computer networks. Basic concept.	i, ii, iii
5	An introduction to computer networks: OSI (7 layers) reference model. The (5 layers) internet model. The (3 layers) EPA Model.	i, ii, iii, iv
6	Physical layer: Data and signals. Analog transmissions.	iii, iv, v
7	Physical layer: Digital transmissions.	iii, iv, v
8	Physical layer: Transmission media. Switching. FDM and TDM.	iii, iv, v
9	Data link layer: Error detection and correction.	iii, iv, v
10	Data link layer: Data link control and protocols. Multiple Access.	iii, iv, v
11	Network layer.	iii, iv, v
12	Transport layer.	iii, iv, v
13	Application layer.	iii, iv, v
14	IEDs and Computer networks for power systems applications.	iv, v

Relationship between the Course and Electrical Engineering Program Curriculum

	Electrical Engineering Program Outcomes	Level of Contribution		
		1	2	3
a	an ability to apply knowledge of mathematics, science, and engineering			X
b	an ability to design and conduct experiments in electrical engineering, as well as to analyze and interpret data to reach an appropriate conclusion		X	
c	an ability to design an electrical system, component, or process to meet desired needs			X
d	an ability to function on multi-disciplinary teams			X
e	an ability to identify, formulate, and solve electrical engineering problems			X
f	an understanding of professional and ethical responsibility		X	
g	an ability to communicate effectively in both oral and written fashion		X	
h	the broad education necessary to understand the impact of engineering solutions in a global and societal context		X	
i	a recognition of the need for, and an ability to engage in life-long learning		X	
j	a knowledge of contemporary issues and their impact on engineering profession	X		
k	an ability to use the techniques, skills, and modern engineering tools necessary for electrical engineering practice			X
l	a knowledge of applied electronics, computer and information systems to design and analyze complex systems for electrical engineering applications			X

1: weak, 2. Partial, 3. Full